

PEDALSHIELD

Prove the ride. Never the route.

Verified proof that a person physically made a trip — without collecting anywhere they went.

Nobody can prove a trip happened without surveilling the traveller.

Every system that rewards or reports physical movement faces the same fork.

Trust the self-report

A survey, or an app where someone taps “I biked today.”

Cheap, and unverifiable. Nobody knows the real number. Incentives pay out on the honor system, and the data is weak evidence for funding or climate reporting.

Collect location data

Continuous GPS from every participant.

Accurate — and now you hold a database of where your employees or students go. Legal objects, participation drops, and it is a breach waiting to happen.

Pedalshield collapses the fork: the phone does the judging, and only the verdict leaves.

The demand is written into law

Washington requires worksites of 100+ employees to measure drive-alone rates every two years. Oregon and California have their own mandates.

1,000+

WA worksites under the
Commute Trip Reduction mandate

2027

California SB 253 adds employee-commuting
emissions to mandatory disclosure

2 min

the self-reported survey that
produces both numbers today

We don't have to create the demand, and we don't sell one worksite at a time

~4,000 US worksites sit under an active commute mandate today. From 2027, every company over \$1B revenue doing business in California must disclose employee-commuting emissions — a far larger set, reporting the same unverified survey. We reach both through the channels that already serve them: commute-software vendors whose weakest link is verification, and regional transportation programs that run the incentives.

What we built

The rider's phone verifies the ride. A signed verdict leaves. The route never does.

1 Sense

GPS, accelerometer, gyroscope and barometer are read on-device during the ride. Nothing is uploaded.

2 Judge

A physics engine scores the ride against what a bicycle can actually do — speed envelopes, pedaling cadence, road vibration, sensor agreement.

3 Sign

The phone emits a distance and an integrity score, signed by a device key. That is the entire payload.

An open-source test fails the build if any location or sensor field could appear in that payload. The privacy promise is executable, not a policy document.

Why this is hard — and why it is the moat

Anyone can claim a trip. Making the claim expensive to fake is the entire product.

Single-sensor checks fall to single-sensor fakes

Replay a real GPS track while shaking the phone, and every conventional check passes.

Wrong answers cost more than missed ones

A rejected honest rider is churn. Impossible stretches are excised and earn nothing; the rest of the ride still pays.

So we check that the sensors agree

The gyroscope must rotate when GPS says you turned. Road vibration must rise as you accelerate. Faking one stream is easy; faking two that stay coherent in time is a different problem.

The engine is the licensable asset

The privacy contract is open source. The scoring engine is not. Its value compounds with real ride data — the part a competitor cannot copy.

It works, on mainnet, with no human in the loop

Every payout is built, proved, signed and broadcast autonomously. Verify any of these yourself.

fbf4e134...d16ed8

First payout after the Zcash hard fork — a cross-pool migration spend

c42a0097...5f72b8

3.07 verified miles · integrity 0.71 · 30,000 zatoshi

7bcda2d6...803164

1.04 verified miles · integrity 0.69 · 17,674 zatoshi

4 min → sec

payout latency, after rebuilding
the treasury's chain scanner

0.62

verification threshold — tuned
against real rejected rides

~\$0.09

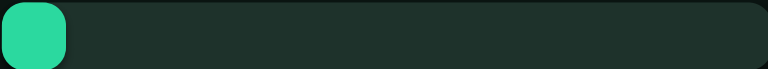
per mile, pegged to the EPA
social cost of carbon

Market — built bottoms-up from a verified anchor

Washington’s 1,000+ mandated worksites sit in counties holding ~6.5M of 340M Americans. Scaled by population: ~50,000 US worksites meet CTR-style criteria.

Mandated worksites today

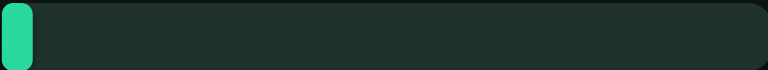
4,000 × \$6k



\$24M

Universities with TDM programs

700 × \$15k



\$11M

CTR-eligible US worksites

50,000 × \$6k



\$300M

\$300M

beachhead TAM, per year

\$6,000 ACV assumes 500 employees at \$12/employee/year — priced as compliance software, because per-trip pricing nets only ~\$1,400 per site.

Not sized here: from 2027 California SB 253 requires every company over \$1B revenue operating in the state to disclose employee-commuting emissions. Same product, separate and larger buyer set — we have not put a number on it because we cannot yet verify the company count.

The primitive is horizontal

Proof someone physically did something, without collecting where. Commute is the wedge, not the ceiling.

1 Commute compliance

~\$300M/yr

Mandated measurement, existing budgets, and no incumbent doing verification. Our beachhead.

2 Usage-based insurance

\$6.9B market in 2026, ~20% of US auto policies and rising to 30–35%.

Built entirely on location harvesting — and four states have introduced bills restricting it. A privacy-preserving layer at 3–5% is \$200–350M/yr.

3 Gig & delivery

DoorDash lost \$2.5M to a single GPS-spoofing scheme, and in 2026 banned a driver for AI-fabricated delivery evidence.

Platforms already buy location-integrity tooling.

Scale comes from licensing the engine into channels that already own the customer — one commute platform reaches thousands of enterprises; one insurance carrier is worth a thousand worksites.

How we make money

Two lines, and neither of them charges the rider.

Compliance software

A per-worksite annual contract. The deliverable is verified trip counts a program manager can put in a statutory report — replacing a self-reported survey they already know is weak.

Engine licensing

The verification engine as an SDK, priced per verified trip or per seat, sold into platforms that already have the customers: commute software, telematics, delivery.

We sell measurement, never offsets.

Buyers get a verified number for their own emissions inventory — not a tradable credit. The rider is already paid the carbon value of the mile, so selling a credit on the same mile would double-count it. Riders never pay, and never sell their data by default; if they opt in to contribute route data, the co-op pays them a share of what it is licensed for.

Founder

Solo. One pair of hands from the Orchard wallet to the App Store build.

Samuel B. Newman

Founder · IntelliGrip Industries · Bend, Oregon

Before this I built eBikes Tubes Direct, an e-commerce tube and tire company I scaled to \$300K — selling to cyclists, one order at a time.

A verified bicycle mile is worth nothing today. Neither is the pollution it avoided. Not because the value isn't real, but because nobody can prove the mile happened.

I am building that proof.

10 weeks

from first commit to a live iOS app paying on mainnet

100

commits, one author

~40k

lines of Rust and TypeScript, shipped

\$300K

revenue at my previous company, also solo

Built solo: a hand-rolled Zcash Orchard wallet, an on-device anti-cheat engine, an autonomous payout treasury, and an iOS app — then kept it paying through a consensus hard fork the same day it broke.

Where we actually are

Stated plainly, because the gap is the point.

Built and live

iOS app shipping on TestFlight

Autonomous shielded payouts on mainnet, no operator

An anti-cheat engine that survived a consensus hard fork

Published threat model, MIT privacy contract

Not yet

One rider — the founder

Zero revenue, zero customers

Thresholds tuned against synthetic attacks and a handful of real rides

No pilot signed

The ask: ten riders and one signed pilot in Bend.

That pilot answers the only question no amount of engineering can: how far apart are self-reported and verified trip counts — and will an organization pay to know.